HW1:

求下列程式執行結果

```cpp
#include <iostream>
#include <string>

struct IMYourGrandFather {
    std::string name;

    IMYourGrandFather(const std::string& name)
        : name{name} {

        std::cout << "I'm your father's father: " << name << "!!" << std::endl;
    }
};

struct IMYourFatherA : public IMYourGrandFather {
    IMYourFatherA()
        : IMYourGrandFather{"Amanda"} {

        std::cout << "I'm your father A!!" << std::endl;
    }
};

struct IMYourFatherB : public IMYourGrandFather {
    IMYourFatherB()
        : IMYourGrandFather{"Boruto"} {

        std::cout << "I'm your father B!!" << std::endl;
    }
};

struct ItsMe
    : public IMYourFatherB
    , public IMYourFatherA {

    ItsMe() {
        std::cout << "It's me, Mario!!" << std::endl;

        std::cout << IMYourFatherA::name << std::endl;
        std::cout << IMYourFatherB::name << std::endl;
    }
};

void
run_this_function() {
    ItsMe mario;
}

int main()
{
    run_this_function();
    std::cout << "Hello, world!\n";
}
```

I'm your father's father: Boruto !!

I'm your father B!!

I'm your father's father: Amanda !!

I'm your father A!!

It's me, Mario!!

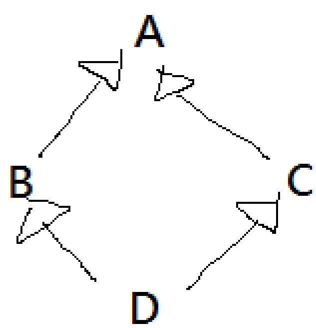
Amanda

Boruto

Hello, world!

HW2:

我如果想要進行鑽石型的繼承 (diamond-shaped inheritance) in C++



請寫下 C++ 的class A,B,C,D 的宣告，來達到這個繼承。

```
class A {
public:
   int x;
};
class B : virtual public A {};
class C : virtual public A {};
class D : public B, public C {};
```